

Narrative Framing, Online Toxicity, and Social Spillover: A Reddit Analysis of the Israeli–Palestinian Conflict

Ettore Miglioranza (257331), Andrea Battaglia (257227), Tommaso Cestari (255480)

Department of Sociology and Social Research, University of Trento

Abstract

This study investigates how the Israeli–Palestinian conflict influences the spread and intensity of toxic discourse online. Using Reddit as a case study, the analysis combines transformer-based linguistic modelling (Detoxify) with dimensionality reduction through principal component analysis (PCA) to examine how toxicity manifests itself in different narrative contexts (military, humanitarian and political) and how it can spill over into apolitical contexts. The results show that humanitarian narratives primarily trigger widespread offensive and vulgar language, political discussions are more associated with explicit threats, and both humanitarian and military contexts reinforce identity-based hostility. Furthermore, longitudinal analysis from 2015 to 2025 reveals a marked increase in toxicity in sports discussions referring to Israel, suggesting that geopolitical hostility can extend to seemingly neutral digital environments. These findings highlight the contextual and structural complexity of online toxicity and show how external crises can reshape patterns of interaction and expression in online communities.

Keywords: *Online Toxicity, Israeli–Palestinian Conflict, Transformer-based Classification Model, Principal Component Analysis (PCA), Narrative Framing, Thematic spillover*

1 Introduction

In recent years, the growing centrality of social media in public debate has highlighted how major geopolitical events can significantly influence the spread of toxic online communications. In this context, the Israeli–Palestinian conflict offers a particularly significant framework for examining how military, political or humanitarian events can trigger the proliferation of aggressive and humiliating content, which often goes beyond the boundaries of political debate and degenerates into broader forms of hostility (Olteanu et al., 2018). This particular conflict has its roots in a long history of territorial, religious and identity tensions, which intensified during the 20th century and have been cyclically reactivated by new escalations (Bartov, 2023). The operations have particularly affected densely populated areas, causing a serious and prolonged humanitarian crisis. The hostilities have had a severe impact on civilians, infrastructure and essential services, worsening living conditions and attracting international public attention (Jamalud-dine et al., 2025).

In this context, the outrage and polarisation generated by the conflict have often fuelled toxic digital narratives, ranging from hostility motivated by ethnic or religious reasons to more general expressions of aggression, dehumanisation and verbal abuse. Such content often exceeds the bounds of legitimate political criticism, fuelling online environments filled with antagonism, misinformation, and emotionally charged language (Goel et al., 2023). One of the most important characteristics of this phenomenon is its ability to extend beyond discussions directly related to the conflict. Toxic narratives triggered by conflict-related events can emerge in contexts seemingly unrelated to politics, such as sport, entertainment or other areas of popular culture. This dynamic, which could be interpreted as a form of “spillover”, suggests that once activated, toxic discourse tends to move transversally across platforms, reaching environments that were not originally related to the geopolitical event. In these spaces, the emotions triggered by violent episodes can alter the tone of discussions, even on unrelated topics, reinforcing antagonistic and dehumanising narra-

tives. If confirmed, this would indicate that such toxicity is not merely a reaction to specific policy choices or political actors, but rather a manifestation of a broader, generalised form of hostility directed towards an entire people, which adapts flexibly to different digital environments and conversational contexts (Schmitz et al., 2022).

Based on these observations, this research aims to explore how public events related to the Israeli–Palestinian conflict influence the intensity and structure of toxic online discourse. Rather than establishing rigid causal relationships, the study investigates how conflict-related events are discursively framed and how these framings influence the forms, registers, and amplification of toxic narratives over time.

The paper will start by conducting a brief analysis of the literature background behind online toxic discourse and geopolitical spillover effects, through the lenses of computational social science methodologies and theories. Following that, the specific context of the recent escalation and the datasets utilized will be introduced, in order to proceed with the project design and the core analyses of this paper. Finally, results will be accurately interpreted in order to draw informative conclusions.

2 Theoretical Background and Literature Review

2.1 Operational Definitions and Conceptual Distinctions in Online Toxicity

The study of online toxicity requires, as a first step, a clear operational definition of the phenomenon. Compared to traditional media, hostile expressions in digital environments take specific forms, exploiting implicit cultural codes, the potential of platforms and algorithmic amplification. Toxicity is generally understood as a broader category that includes hate speech, but also encompasses insults, harassment, threats, and other forms of verbal abuse not necessarily based on group identity (Wulczyn et al., 2017). A widely cited working definition of hate speech, as a subset of toxic communication, comes from the Council of Europe, which identifies it as all forms of expression ‘which spread, in-

cite, promote or justify racial hatred, xenophobia, anti-Semitism or other forms of hatred based on intolerance, including intolerance expressed by aggressive nationalism and ethnocentrism, discrimination and hostility against minorities, migrants and people of immigrant origin' (Council of Europe, 1997). More recently, the United Nations has emphasised a similar perspective, defining hate speech as 'any kind of communication in speech, writing or behaviour, that attacks or uses pejorative or discriminatory language with reference to a person or a group on the basis of who they are' (United Nations, 2019).

One of the main difficulties lies in distinguishing emotionally charged but legitimate political expression from offensive communication. Much hostile content takes the form of irony, ambiguous metaphors or seemingly generic rhetoric, which complicates classification. Automatic detection systems often struggle to handle such pragmatic ambiguity (Vidgen et al., 2019), while discursive context plays a decisive role in interpretation (Talat et al., 2018). For example, a seemingly neutral statement can take on a toxic meaning depending on the conversational context or community norms. Scholars have therefore proposed various criteria for separating offensive or hateful communication from legitimate criticism or dissent. Steffen et al., 2024 suggest focusing on intent and argumentative structure: hate speech is characterised by attacks on collective identity, stereotypes and dehumanisation, while political criticism tends to be directed at institutional decisions or policy choices, using arguments based on concrete evidence. Vidgen et al., 2020 emphasise the need for context-sensitive approaches that integrate linguistic signals with socio-political information to reduce classification errors.

From this perspective, specific manifestations of hatred towards certain groups, such as anti-Semitism, Islamophobia or misogyny, should be understood as particular examples of toxic communication. Each has historical, cultural and symbolic dimensions that make it recognisable in context, but they all share common structural characteristics: toxicity as a communicative practice aimed at exclusion, stigmatisation and hostility. This broader conceptualisation also allows researchers to capture offensive discourse that does not fall within the stricter definitions of hate, but which nonetheless contributes to the deterioration of online discussions and civil life (Hosseinmardi et al., 2021).

2.2 Computational Approaches and Empirical Analyses

In addition to theoretical reflections, a growing number of studies have addressed the issue of online toxicity from an empirical and computational perspective, producing annotated datasets, classification models and benchmarks. One of the oldest and most widely used resources is the Hate Speech and Offensive Language dataset by Davidson et al., 2017, which distinguishes between hateful content, offensive but non-hateful content, and neutral expressions on Twitter. Building on this, Founta et al., 2018 expanded the approach by collecting a large-scale dataset of abusive behaviour, using crowdsourcing to capture different categories of harmful speech. Similarly, Zampieri et al., 2019 introduced the Offensive Language Identification dataset within the SemEval shared tasks, which remains a benchmark for evaluating computational models in multilingual contexts.

These datasets have supported the development of increasingly sophisticated classification models. Previous approaches relied on shallow methods such as logistic regression or support vector ma-

chines, often trained on bag-of-words or n-gram representations. More recent studies demonstrate the superiority of transformer-based architectures, such as BERT and RoBERTa, in detecting subtle or implicit forms of toxic communication (Vidgen et al., 2020). Mozafari et al., 2020 demonstrate that BERT-based models not only achieve higher accuracy than traditional machine learning approaches, but also mitigate classification bias, while Liu et al., 2019 demonstrate the advantages of transfer learning for detecting multilingual offensive language in shared benchmark tasks. These results confirm that newer architectures are better suited to capturing pragmatic ambiguity and context dependence, which are frequent characteristics of online hostility.

In addition to supervised classification, researchers employed statistical techniques such as factor analysis to uncover the latent dimensions of toxic discourse. Principal component analysis (PCA) is often used as a dimensionality reduction method, allowing related indicators of hostility (e.g., toxicity, insults, threats) to be grouped into broader components that capture underlying patterns of abuse. By quantifying these latent structures, PCA allows us to interpret how different forms of toxicity coexist and to compare their relative prevalence in different contexts. This approach has proven useful for complementing classification models, as it shifts the focus from detecting individual categories to analysing the structural relationships between them. Recent applications of PCA and related factor analysis methods to online communication data show how these techniques can uncover latent patterns and underlying structures in user behavior or discourse that may not be captured by classification models alone (Tien et al., 2020).

2.3 Diffusion of Online Toxicity and Platform Governance

While computational approaches have focused primarily on classification and detection, a complementary line of research has analysed the dynamics of online toxicity, highlighting its spread, escalation and transformation across platforms. Mathew et al., 2019 demonstrate that in order to assess the social impact of toxic discourse, it is essential to measure not only its prevalence but also its virality within communities, as highly shared hostile content can shape perceptions and normalise antagonism.

Echo chambers play a decisive role in these processes. Goel et al., 2023 demonstrate how ideologically cohesive groups contribute disproportionately to the spread of toxic narratives, reinforcing hostility through closed communication circuits. Schmitz et al., 2022 expand on this perspective by demonstrating that participation in toxic communities can radicalise users, with effects observable beyond the boundaries of such groups. This indicates that online hostility is not static, but can expand, intensify, and spill over into mainstream discourse.

The influence of external crises and geopolitical conflicts further amplifies these mechanisms. Kumar et al., 2018 reveal how user sentiment is strongly influenced by the tone of mainstream news posts, while He et al., 2021 document how the COVID-19 pandemic triggered waves of toxic and counter-toxic communication, illustrating how the emotional climate of one context can permeate other contexts. These studies highlight that online toxicity often transcends the specific event that generated it, potentially emerging in spaces seemingly unrelated to politics, such as discussions about sport or entertainment.

A concrete example of how platforms attempt to manage these dynamics is provided by Reddit's policy on toxic content. Since

2020, Reddit has explicitly banned content that “promotes hatred based on identity or vulnerability”, applying sanctions not only to individual posts but also to entire communities (Reddit, 2020). Research shows that banning toxic subreddits significantly reduces exposure to harmful content, although displaced users may migrate to other platforms (Chandrasekharan et al., 2017; Jhaver et al., 2019). This illustrates both the potential and limitations of platform governance in addressing spillover effects and radicalisation.

3 Research Questions

The emergence and spread of online toxicity on social media is a complex phenomenon, strongly influenced by the socio-political context in which it develops. This study takes the Israeli–Palestinian conflict as a case study to explore how episodes of violence, military operations or humanitarian crises resonate in digital arenas and influence the ways in which hostility is expressed. The focus is on two main dimensions: how the conflict is framed in online debates and how the toxic reactions triggered in these discussions can extend beyond the political sphere to other thematic areas.

Consequently, the research is organised around two guiding questions:

- **RQ1 – Narrative framing and toxic reactions:** How does the way in which conflict-related events are framed (e.g., military, humanitarian, political) influence the tone and intensity of toxic discourse in online spaces?
- **RQ2 – Spillover into apolitical contexts:** To what extent does toxicity related to the Israeli–Palestinian conflict spread into online discussions that are not directly political, such as those concerning sports or cultural topics?

By addressing these questions, the study aims to contribute to a more nuanced understanding of how online toxicity operates within and across digital environments, and to assess the extent to which toxic reactions triggered by geopolitical events remain confined to political discourse or spread to broader areas of public discourse.

4 Methodology

4.1 Data Collection and Preprocessing

The data used in this study was retrieved from Reddit using Reddit’s official API. Using a configuration file specifying the subreddits, search categories, and relevant time window, the scraping script was designed to retrieve posts that matched a predefined set of queries. For each retrieved post, the complete comment tree was downloaded and analysed recursively, ensuring that the entire discussion thread related to each topic was captured. Only comments with a minimum positive score were retained for analysis in order to exclude irrelevant or low-quality content.

The collection covered two time periods. The first focused on the period of conflict (18–25 March 2025), corresponding to a phase of intense military operations in Gaza. The second extended to a longitudinal period of fallout (January 2015 – August 2025), designed to observe how hostility related to the conflict could emerge in discussions that were not explicitly political. In total, four thematic categories were used to structure the collection of Reddit content, corresponding to distinct narrative and discursive contexts.

The first category is “Israeli military actions” and includes posts referring to direct military operations – air strikes, bombings, ground incursions and missile attacks – which represent the most concrete and operationally focused form of coverage of the conflict. The queries (e.g., “raid on Rafah”, “Israeli air strike on Gaza”) were designed to capture discussions focused on tactical or battlefield events, without explicit humanitarian or political reference.

The second category (“Humanitarian Narrative”) targeted content describing the human consequences of the conflict, such as civilian casualties, attacks on hospitals, and displacement crises. Some examples are “UNICEF Gaza” and “Gaza famine.” The goal was to identify discussions that emphasised civilian suffering and moral condemnation, distinguishing them from strictly military or strategic discourse.

The third (“Political Representation”) included posts discussing official reactions, institutional statements, and international diplomacy (e.g., “UN condemns Israel”, “Netanyahu negotiates ceasefire”). These queries were designed to capture the framing of the conflict as a political and diplomatic issue, allowing for comparison with more emotionally charged humanitarian and military narratives.

Finally, to investigate the potential spillover of toxic narratives into non-political environments, an additional set of queries targeted discussions in subreddits dedicated to sports and entertainment (e.g., r/soccer, r/Music, r/movies, r/television, r/popculturechat). The selected queries focused on public figures identified as Jewish or Israeli. The inclusion of this category made it possible to identify possible toxic discourse emerging in seemingly neutral contexts, where the mere reference to identity could trigger polarised or hostile reactions.

A more complex Boolean search query was used than for the other categories, due to the significantly higher volume of content produced in the main subreddits dedicated to sport and culture. The query was structured using compound filters that required (a) a reference to Israel or Israeli identity in the post title (e.g., “Israel”, “Israeli”, “ISR”), (b) the co-occurrence of terms explicitly related to sport such as player, athlete, coach, team or league, and (c) the exclusion of high-volume, low-signal routine content such as match threads, game threads, previews and summaries. An example of one such structured query is as follows: ‘title:(Israel OR Israeli OR ISR) (title:player OR title:athlete OR title:coach OR title:team OR title:club OR title:league OR title:academy) -’match discussion’ -’game discussion’ -preview -summary’. Each comment was stored along with its metadata, including post ID, title, publication time, positive votes, and assigned category.

Overall, this strategy yielded a corpus of roughly 20,000 Reddit comments reflecting multiple narrative frameworks of the Israeli–Palestinian conflict, alongside apolitical discussions with potential identity-based tensions. By organizing data collection in both conflict-related and neutral contexts, the study aimed to capture a bigger part of the spectrum of online discourse, from explicitly political debates to subtle manifestations of hostility in everyday digital life.

Figure 1 shows the daily volume of comments retrieved for each narrative frame between March 18 and 25, 2025. This temporal window was chosen because on March 18, the ceasefire that had been in place since January collapsed when Israel launched a large-scale attack on Gaza, including intensive airstrikes that killed hundreds of people and caused the resumption of the war. The tempo-

ral pattern corresponds to the typical dynamics observed in online discussions following major geopolitical or violent events. Immediately after the intensification of military operations in the Gaza Strip (March 18–19), the categories “Israeli military actions” and “political representation” show a significant increase. This concentration reflects the initial focus of public attention on the actual course of the attacks and the reactions of political institutions, which typically dominate the early stages of information flow in the context of crises.

However, the distribution shifts over the course of the days. Comments related to the humanitarian situation increase significantly from March 22 and become the most prominent category by March 24. This pattern is consistent with the fact that humanitarian consequences—civilian casualties, damage to infrastructure, and reports from aid organizations—usually only become more visible once the immediate military developments have been reported.

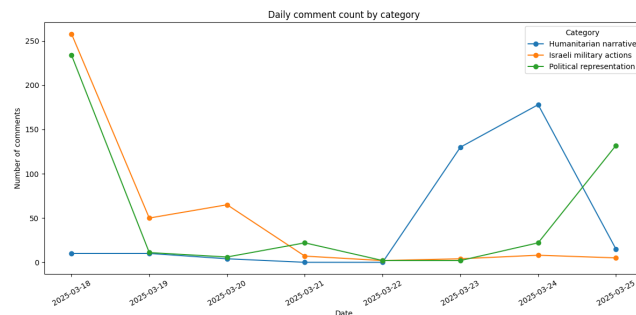


Figure 1. Daily comment count by category during the conflict period (March 18–25, 2025).

Figure 2 shows the annual volume of comments retrieved in apolitical contexts referring to Israel, distinguishing between cultural and sports discussions. Throughout most of the 2015–2025 period, sports-related threads consistently generate significantly more comments than cultural ones, with particularly pronounced peaks in 2016, 2018, 2021, and 2023. This persistent gap is consistent with the structural characteristics of online platforms: sports-dedicated subreddits, particularly large communities such as r/soccer, r/nba, and r/cycling, generate much higher baseline engagement and conversation activity than cultural spaces, resulting in a greater number of discussions where references to Israeli athletes, clubs, or teams may emerge.

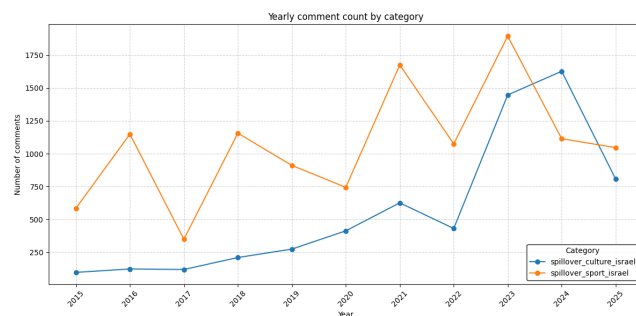


Figure 2. Yearly comment count by category for spillover contexts, 2015–2025.

4.2 Content Classification Model with Detoxify

The Reddit comments were classified using Detoxify, an open-source Python library that provides transformer-based models for detecting toxic language. Detoxify is based on the RoBERTa architecture and was trained using the Jigsaw Toxic Comment Classification Challenge dataset, which includes millions of commented posts from online platforms. This challenge was created specifically in response to the observation that previous models for detecting toxic language exhibited systemic bias and often misclassified comments containing neutral identity terms (such as race, gender, or sexual orientation) as toxic by associating them with toxic language. For each comment considered, the model outputs probabilities for six categories of abusive language: toxic, highly toxic, obscene, threatening, insulting, and identity-hostile. Each rating represents the estimated probability that a given comment contains that specific form of hostility.

Annotators were asked to apply the following definitions:

- **Toxic:** rude, disrespectful, or unreasonable comments.
- **Severely toxic:** extremely hateful, aggressive, or offensive content, more severe than “toxic”, including explicit insults or violent threats.
- **Insulting:** derogatory or provocative remarks directed at individuals or groups, including vulgar language and generic insults.
- **Obscene:** vulgar or profane expressions, including sexual language that is not necessarily offensive.
- **Threatening:** explicit statements of intent to cause harm, injury, or violence to others.
- **Identity attack:** Negative or hateful comments that target attributes of personal identity such as race, religion, or gender.

The use of continuous outputs rather than binary labels provides a more nuanced representation of the data. Instead of reducing each comment to a rigid toxic/non-toxic classification, probabilistic scores allow for the capture of varying degrees of hostility and ambiguity in expression.

The unbiased version of Detoxify was used for this study, as it has been retrained on extensive datasets to reduce systematic misclassifications, particularly those involving identity-related terms (e.g., references to nationality, religion, or ethnicity). This choice is crucial in the context of the Israeli–Palestinian conflict, where politically charged discussions often overlap with identity-based expressions that are not necessarily hateful. To better understand the classifier’s behavior in the context of politically and emotionally charged discourse, a small qualitative evaluation was conducted on individual Reddit comments. The goal is not to provide statistical estimates of accuracy, but to illustrate typical patterns of correct detection and misclassification observed in the dataset, particularly in borderline or context-dependent cases.

Table 1
Qualitative evaluation examples of model classification.

Category	Example & Analysis	Model Output
True Positive (TP)	“Bunch of animals. Zionists are not humans. I refuse to believe it.” <i>This comment constitutes an explicit dehumanizing attack directed at a collective identity group. The model correctly assigns high toxicity and identity-attack scores despite the absence of profanity or explicit threats.</i>	toxicity = 0.873 identity_attack = 0.725
True Negative (TN)	“I really just wanna see what would’ve happened if all the hostages were returned from the beginning, what would their excuse be” <i>While this comment implicitly questions the actions and justifications of political actors, it does not employ offensive language, personal attacks, or dehumanizing rhetoric</i>	toxicity = 0.001
False Positive (FP)	“These people have no morals. If 10 innocent Palestinian children get killed in an explosion but 1 Hamas fighter also dies they see it as a win. Ends justify the means and all that shit. It’s despicable really.” <i>While the model correctly flags the presence of vulgar language with a high obscene score, it incorrectly interprets profanity as evidence of a disrespectful comment.</i>	toxicity = 0.959 obscene = 0.84
False Negative (FN)	“Gaza seems to have more hospitals per capita than anywhere else on earth.” <i>Although superficially neutral, this comment appears to employ irony or sarcasm in reference to attacks on medical infrastructure, which the model does not catch.</i>	toxicity = 0.0

Across the set of comments, several systematic patterns emerge. Profanities and vulgar markers substantially increase toxicity scores, even when the profanity is not directed at any group and is incidental to the comment’s meaning. Comments involving geopolitical vocabulary also tend to receive inflated toxicity scores

because such terms frequently co-occur with genuinely hostile discourse in the training data. At the same time, some comments contain sarcasm or rhetorical framing, which models struggle to interpret. This illustrates a broader challenge: transformer-based toxicity classifiers perform poorly when harmful intent is conveyed implicitly rather than through lexically marked hostility. Such cases underscore the difficulty of detecting subtle or coded forms of antagonism in discourse surrounding geopolitical crises.

5 Results

5.1 Narrative Framing and Hostile Reactions

The results relating to the coexistence of different dimensions of hostility provide an initial overview of the structure of toxic discourse in the dataset. As shown in Figure 3, which shows the heat map of correlations, there are strong positive associations between toxicity and offensive and obscene language. This indicates that when users employ a generally toxic tone, they often do so through profanity or direct personal denigration. The correlation between toxicity and insults is particularly high, suggesting that insults are one of the most typical expressions of toxicity in online discussions related to conflict. In contrast, threats show only weak or no correlations with other variables, highlighting their peculiarity: explicit threats are not necessarily accompanied by obscene or offensive language, but represent a separate way of expressing hostility. This preliminary overview already suggests that online abuse is not a homogeneous phenomenon, but rather consists of partially independent registers.

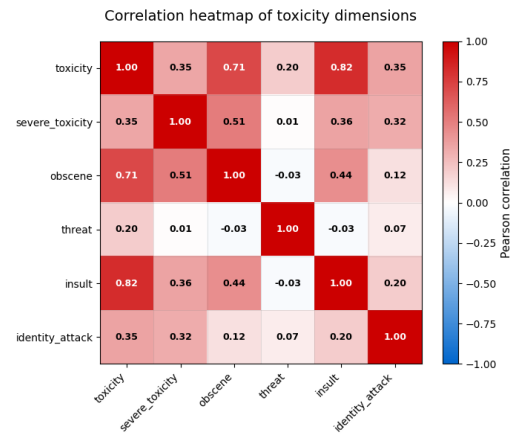


Figure 3. Correlation heatmap of toxicity dimensions.

In order to identify these underlying structures more systematically, a principal component analysis (PCA) was conducted as part of the factor analysis. The first three components, which together explain approximately 80% of the total variance, were retained for interpretation. The loading matrix is shown in Table 2. In line with standard practice, only variables that contribute more than 15% to each component are discussed here, as they define the substantive meaning of the latent dimensions. The first component is characterized by high loadings for toxicity, insults, and obscene language. This combination can be interpreted as a dimension of verbal abuse and vulgarity in a broad sense, in which aggression is expressed in widespread and emotionally charged forms of language. The second component is explained almost entirely by

threats, which clearly represent a distinct register focused on explicit calls for violence or intimidation. The third component, on the other hand, is dominated by attacks on identity and severe toxicity. This structure indicates a dimension of identity-based hostility, in which antagonism is directed towards groups or individuals because of their membership, often through ethnicized and dehumanizing expressions. The PCA therefore confirms that the forms of hostility identified by the model fall into three main categories: widespread verbal abuse, explicit threats, and identity-based attacks.

	PC1	PC2	PC3	PC4	PC5	PC6
toxicity	30.74	0.93	-6.80	-6.03	2.48	53.01
severe_toxicity	15.40	-0.81	17.31	39.13	-24.13	3.21
obscene	22.49	-5.15	-2.93	14.93	38.77	-15.73
threat	0.36	79.94	-8.90	7.59	-0.58	-2.63
insult	23.59	-1.42	-6.70	-18.06	-27.51	-22.72
identity_attack	7.42	11.74	57.36	-14.26	6.53	-2.69
Total variance explained	46.19	17.66	15.48	12.43	7.65	0.58

Table 2

Signed contribution (%) of each toxicity variable to the six principal components (PCA).

The next step is to examine how these latent components are distributed across the three narrative contexts that characterize the conflict-related news in the dataset: humanitarian narrative, political representation, and Israeli military actions. The average scores per category on each principal component are shown in Figures 4–6. It is important to note that the values shown in the bar charts represent normalized scores. A negative value in these figures does not indicate the absence of a particular type of hostility, but rather that the average score for that narrative framework is lower than the overall average across categories. Conversely, positive values indicate an above-average prevalence of that component. This normalization makes the graphs suitable for comparison across categories, but they should be interpreted as relative measures rather than absolute counts.

The results reveal clear differences in how hostility is articulated depending on the framing of the event. In the case of humanitarian narratives, comments tend to focus heavily on the first component, suggesting that news about civilian suffering, famine, or the bombing of hospitals generate intense reactions expressed through vulgar language and verbal insults. The emotional charge of such content seems to mobilize a wave of indignation that manifests itself linguistically in offensive and often vulgar terms, rather than in structured threats or identity-based attacks.

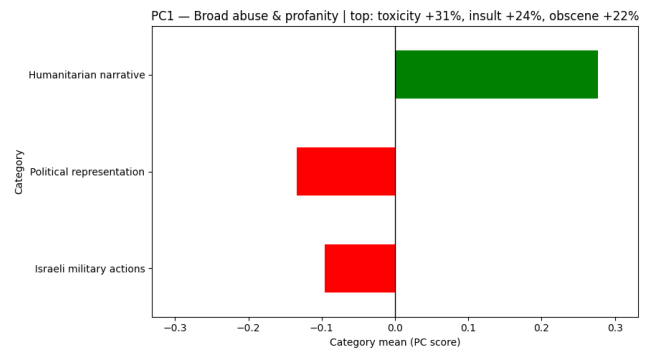


Figure 4. Distribution of narrative frames on PC1 - Broad verbal abuse and profanity.

Political representation, on the other hand, shows a stronger association with the second component. As shown in Figure 5, discussions about political statements, institutional debates, or declarations of war are more frequently accompanied by explicit threats. This indicates that when conflict is framed in political terms, users are more likely to resort to violent language that directly incites aggressive actions, while profanity and identity-based attacks play a secondary role. The political arena thus seems to act as a trigger for more targeted and confrontational forms of hostility, in line with the polarized nature of institutional debates.

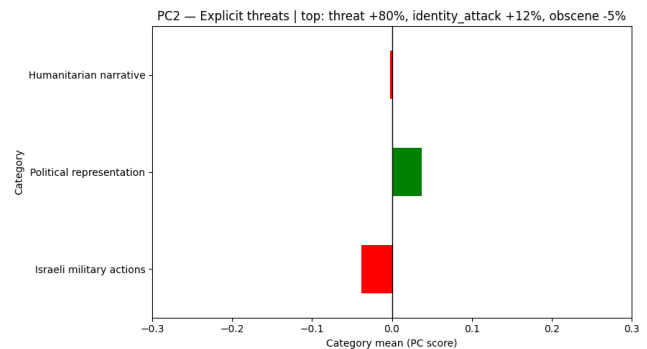


Figure 5. Distribution of narrative frames on PC2 - Explicit threats.

Finally, the third component highlights a different configuration. As shown in Figure 6, both humanitarian narratives and Israeli military actions show positive values on the identity-based hostility dimension, while political representation appears more detached. This suggests that discussions focused on civilian suffering and concrete military operations both tend to generate comments in which antagonism is articulated through identity categories, with users often framing the conflict as a clash between groups rather than as a sequence of strategic or political decisions. Such reactions indicate the persistence of dehumanizing or ethnicized discourses, which are less evident in the more institutionalized context of political debate.

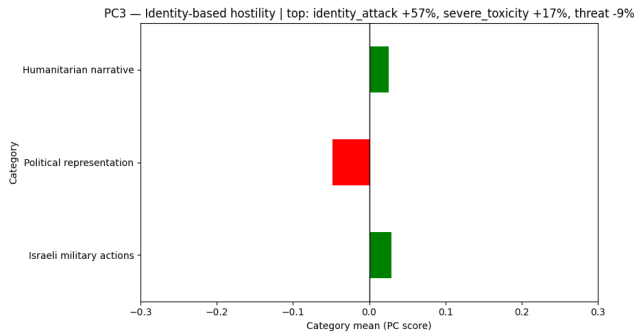


Figure 6. Distribution of narrative frames on PC3 - Identity-based hostility.

Taken together, these findings show that the way conflict-related events are framed in online discussions not only influences the overall intensity of reactions, but also determines the type of hostility expressed. Humanitarian frames mainly mobilize widespread offensive language, political frames are more likely to trigger explicit threats, while both humanitarian and military frames reinforce identity-based hostility. This differentiation highlights the importance of considering narrative framing when analyzing the circulation of hate speech online, as the same conflict can cause very different registers of hostility depending on how it is represented in the digital arena.

5.2 Spillover into Apolitical Contexts

This part of the analysis focuses on Reddit comments responding to apolitical posts that refer to Israel, particularly in the areas of sports and culture. These contexts were selected in order to verify whether hostile language related to the conflict also emerges in discussions that are not explicitly political. As in the previous sections, comments marked as [removed] by Reddit’s moderation system were considered toxic, as their removal reflects a prior assessment of unacceptable or offensive content by the platform.

Figure 7 presents the annual average values for each of the six toxicity dimensions between 2015 and 2025. Throughout the entire period, comments related to sports show consistently higher averages than those related to culture. This difference is particularly marked with regard to toxicity, insults, and obscene language, which show a clear separation between the two categories. Fluctuations over time are observed in both categories, but the overall gap between sports and cultural discussions persists throughout the decade. The graphs also include comments classified as toxic and removed by moderators, indicating that a significant portion of the discourse in these apolitical contexts was deemed offensive directly by the platform.

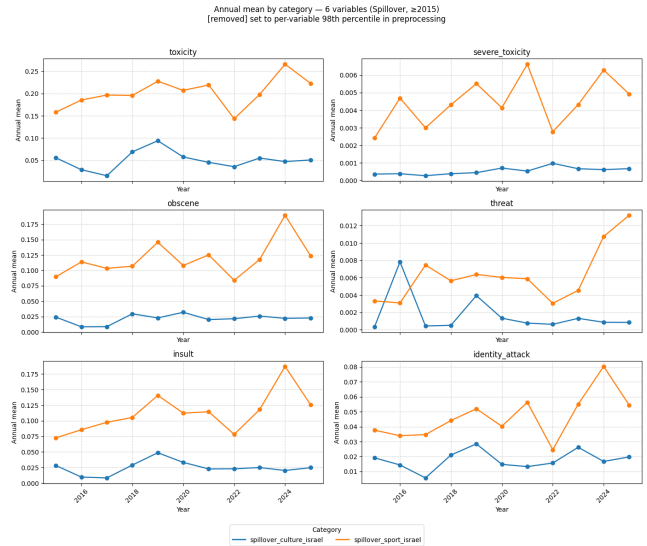


Figure 7. Annual mean values of six toxicity dimensions in apolitical posts related to Israel (sport vs. culture), 2015–2025. Comments flagged as [removed] were treated as toxic, reflecting Reddit moderation decisions.

To summarize these dynamics, a composite toxicity index was calculated by averaging the six variables after normalization. The results are shown in Figure 8. The left panel shows the annual averages of the composite index, confirming that discussions related to sports consistently produce higher values than those related to cultural topics. The right panel illustrates the same measure calculated as a three-year moving average, which smooths out short-term fluctuations and highlights long-term trends. Although both categories show relative stability during the first part of the observation period, the sports category shows a clear increase in recent years. This increase in toxicity in discussions related to sports seems to coincide with the resurgence of the Israeli-Palestinian conflict, suggesting that episodes of intensified violence and international visibility may have been reflected in the tone of apolitical conversations.

In contrast, cultural discussions remain relatively stable over the same period, with no equivalent increase in either annual averages or moving average trends. This divergence indicates that the apparent increase in toxicity is not generalized across all apolitical domains, but appears to be concentrated in the sporting context. Other communities and thematic spaces not directly focused on sports do not show the same trajectory, reinforcing the impression that the spread of toxic discourse is uneven across different apolitical environments.

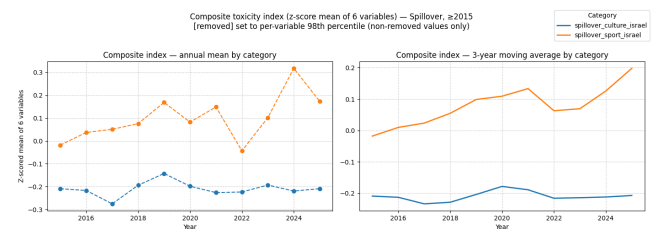


Figure 8. Composite toxicity index (z-score mean of six variables) for apolitical posts related to Israel. Left: annual averages. Right: three-year moving average.

6 Conclusions

The analysis reveals that the expression of hostility in online discussions about the Israeli–Palestinian conflict is not uniform, but depends heavily on how events are represented. Distinct narrative contexts correspond to specific configurations of toxic expressions. When the conflict is described in humanitarian terms, focusing on civilian suffering or moral outrage, users tend to respond with emotionally charged language, rich in profanity and generic insults. These reactions, although intense, are largely indirect and reflect an outburst of frustration rather than targeted aggression. In contrast, discussions framed in institutional or diplomatic language are more prone to explicit threats and calls for violent action, suggesting that political argumentation can serve as a catalyst for conflictual rhetoric. Finally, threads focused on military operations and humanitarian emergencies share a tendency toward identity-centered antagonism, in which users express hostility through collective categories such as ethnicity or religion. Taken together, these patterns indicate that the framing of a crisis mediates not only the volume but also the register of online toxicity, shaping the way aggression is linguistically enacted in digital debate.

Beyond the confines of conflict-related discussions, the longitudinal component of the study provides evidence that hostility linked to geopolitical crises can reverberate into unrelated communicative domains. The gradual increase in toxic content detected in sports-oriented Reddit communities, in contrast to the relative stability of cultural discussions, suggests that antagonistic discourse does not remain confined to explicitly political arenas. Rather, it can infiltrate everyday conversations where identity signals evoke latent tensions. This asymmetric diffusion implies that the emotional charge associated with a distant conflict can permeate seemingly neutral spaces, reshaping the way users interact even in contexts detached from politics.

6.1 Critical analysis

The findings of this paper advance the current understanding of online toxicity in three ways. First, they empirically demonstrate that toxic communication is not a monolithic phenomenon, but a set of differentiated modes of expression structured by a narrative framework. Second, they provide quantitative evidence of propagation across different domains, supporting the idea that digital hostility can migrate across thematic boundaries. Third, the study is consistent with recent computational social research that combines transformer-based linguistic models with statistical methods such as factor analysis to explore the underlying dimensions of online aggression.

Overall, the study’s findings support existing research on digital hostility by emphasizing its context dependence and potential to extend beyond the original arena of conflict, highlighting the value of more nuanced analyses of how online discourse responds to external crises.

The results of this study must be considered within the context of its limitations. A primary limitation is its reliance on publicly available data after platform moderation. Comments removed from Reddit, which likely represent the most severe cases of toxicity, were partially considered but could not be fully analyzed, particularly in the PCA. This inherent sampling bias suggests that our findings on the prevalence and intensity of toxicity are likely

a conservative estimate of the actual phenomenon.

Further limitations arise from the data collection methodology itself. Posts were assigned to “military”, “humanitarian”, or “political” contexts based on the search queries used for retrieval. This method assumes that the resulting comment threads remain thematically consistent with the initial query. However, in practice, these categories are not mutually exclusive; a single post found via a ‘military’ keyword (e.g., ‘Rafah raid’) often contains extensive discussion of its ‘humanitarian’ impact. This overlap at the comment level acts as a confounding variable, meaning that the distinct toxic patterns identified by PCA for each category may be less clearly separated than the analysis suggests.

Another simplification concerns the longitudinal analysis of spillover effects. By aggregating multiple subreddits into broad categories such as “sports” and “culture”, the analysis failed to account for the substantial heterogeneity among communities, not only in terms of moderation norms and practices, but also in the composition and behavioral tendencies of their user bases. Different subreddits attract distinct audiences with varying levels of polarization, engagement styles, and tolerance for conflictual discourse (Chandrasekharan et al. 2017; Jhaver et al. 2021). Consequently, the observed increase in toxicity in the “sports” category may reflect the dynamics of certain influential or particularly active communities rather than a pattern that can be generalized to all sports-related discussions.

Finally, although this study measured the presence of toxic comments, it did not fully account for their social relevance or level of community approval. During data collection, a minimum threshold of positive votes was applied, thus excluding comments that did not receive any positive reception, but this procedure provided only an approximate filter. Among the comments that passed this threshold, variations in the number of positive votes were not incorporated as weighting factors in the analysis. Future work could adopt score-weighted measures of toxicity to distinguish between hostile expressions that are marginal or rejected by the community and those that are widely approved and amplified, offering a clearer understanding of their social impact.

7 Additional Resources

The complete code used to develop the statistical analyses employed in this paper is freely available and can be found at the following GitHub repository¹ for replicability purposes.

References

- Barav, Omer (2023). *Genocide, the Holocaust and Israel–Palestine: First-Person History in Times of Crisis*. en. Bloomsbury Academic. ISBN: 9781350332317. DOI: 10.5040/9781350293144. URL: <https://www.torrossa.com/en/resources/an/5605689>.
- Chandrasekharan, Eshwar et al. (Dec. 2017). “You Can’t Stay Here: The Efficacy of Reddit’s 2015 Ban Examined Through Hate Speech”. en. In: *Proceedings of the ACM on Human-Computer Interaction* 1.CSCW, pp. 1–22. ISSN: 2573-0142. DOI: 10.1145/3134666. URL: <https://doi.org/10.1145/3134666> (visited on 12/27/2025).
- Council of Europe (Oct. 1997). “Recommendation No. R (97) 20 of the Committee of Ministers to Member States on “Hate Speech””. en. In: Strasbourg: Council of Europe. URL: <https://rm.coe.int/1680505d5b> (visited on 12/27/2025).
- Davidson, Thomas et al. (2017). “Automated hate speech detection and the problem of offensive language”. en. In: *Proceedings of the 11th International AAAI Conference on Web and Social Media* 11.1, pp. 512–515. DOI: 10.1609/icwsm.v11i1.14955. URL: <https://doi.org/10.1609/icwsm.v11i1.14955> (visited on 12/27/2025).

¹ <https://github.com/ettoremiglioranza1012/NLP-for-CSS.git>

- Founta, Antigoni et al. (2018). "Large Scale Crowdsourcing and Characterization of Twitter Abusive Behavior". en. In: *Proceedings of the International AAAI Conference on Web and Social Media* 12.1. DOI: 10.1609/icwsm.v12i1.14991. URL: <https://doi.org/10.1609/icwsm.v12i1.14991>.
- Goel, Vasu et al. (2023). "Hatemongers ride on echo chambers to escalate hate speech diffusion". en. In: *PNAS Nexus* 2.3, pgad041. DOI: 10.1093/pnasnexus/pgad041. URL: <https://doi.org/10.1093/pnasnexus/pgad041>.
- He, Bing et al. (2021). "Racism is a virus: Anti-Asian hate and counterspeech in social media during the COVID-19 crisis". en. In: *Proceedings of the IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pp. 90–94. DOI: 10.1145/3487351.3488324. URL: <https://doi.org/10.1145/3487351.3488324>.
- Hosseinnardi, Homa et al. (2021). "Examining the consumption of radical content on YouTube". en. In: *Proceedings of the National Academy of Sciences* 118.32, e2101967118. DOI: 10.1073/pnas.2101967118. URL: <https://doi.org/10.1073/pnas.2101967118>.
- Jamaluddine, Zeina et al. (Jan. 2025). "Traumatic injury mortality in the Gaza Strip from Oct 7, 2023, to June 30, 2024: a capture–recapture analysis". en. In: *The Lancet* 405.10472, pp. 135–143. ISSN: 0140-6736. DOI: 10.1016/S0140-6736(24)02081-0. URL: [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(24\)02678-3/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(24)02678-3/fulltext) (visited on 12/27/2025).
- Jhaver, Shagun, Amy Bruckman, and Eric Gilbert (2019). "Does Transparency in Moderation Really Matter?: User Behavior After Content Removal Explanations on Reddit". en. In: *Proceedings of the ACM on Human-Computer Interaction* 3.CSCW, pp. 1–27. DOI: 10.1145/3359252. URL: <https://doi.org/10.1145/3359252>.
- Kumar, Srijan et al. (2018). "Community interaction and conflict on the web". en. In: *Proceedings of the 2018 World Wide Web Conference (WWW 2018)*, pp. 933–943. DOI: 10.1145/3178876.3186141. URL: <https://doi.org/10.1145/3178876.3186141> (visited on 12/27/2025).
- Liu, Ping, Wen Li, and Liang Zou (June 2019). "NULI at SemEval-2019 Task 6: Transfer learning in multilingual offensive language detection". en. In: *Proceedings of the 13th International Workshop on Semantic Evaluation (SemEval 2019)*, pp. 87–91. DOI: 10.18653/v1/S19-2011. URL: <https://aclanthology.org/S19-2011/> (visited on 12/27/2025).
- Mathew, Binny et al. (2019). "Thou Shalt Not Hate: Countering Online Hate Speech". en. In: *Proceedings of the International AAAI Conference on Web and Social Media* 13.1, pp. 369–380. DOI: 10.1609/icwsm.v13i01.3237. URL: <https://doi.org/10.1609/icwsm.v13i01.3237>.
- Mozafari, Marzieh, Reza Farahbakhsh, and Noel Crespi (2020). "Hate speech detection and racial bias mitigation in social media based on BERT model". en. In: *PLOS ONE* 15.8, e0237861. DOI: 10.1371/journal.pone.0237861. URL: <https://doi.org/10.1371/journal.pone.0237861>.
- Olteanu, Alexandra et al. (2018). "The Effect of Extremist Violence on Hateful Speech Online". en. In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 12. 1, pp. 668–671. DOI: 10.1609/icwsm.v12i1.15016. URL: <https://doi.org/10.1609/icwsm.v12i1.15016>.
- Reddit (2020). *Promoting Hate Based on Identity or Vulnerability*. en. URL: <https://support.reddithelp.com/hc/en-us/articles/360045715951-Promoting-Hate-Based-on-Identity-or-Vulnerability>.
- Schmitz, Matheus, Goran Muric, and Keith Burghardt (2022). "Quantifying How Hateful Communities Radicalize Online Users". en. In: *Proceedings of the IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining*. DOI: 10.1109/ASONAM55673.2022.10068644. URL: <https://doi.org/10.1109/ASONAM55673.2022.10068644>.
- Steffen, Elisabeth, Milena Pustet, and Helena Mihaljević (2024). "Algorithms Against Antisemitism? Towards the Automated Detection of Antisemitic Content Online". en. In: *Antisemitism in Online Communication: Transdisciplinary Approaches to Hate Speech in the Twenty-First Century*. Open Book Publishers, pp. 205–236. DOI: 10.11647/obp.0406.08. URL: <https://doi.org/10.11647/obp.0406.08>.
- Talat, Zeerak, James Thorne, and Joachim Bingel (2018). "Bridging the Gaps: Multi-task Learning for Domain Transfer of Hate Speech Detection". en. In: *Online Harassment*. Springer, pp. 29–55. DOI: 10.1007/978-3-319-78583-7_3. URL: https://doi.org/10.1007/978-3-319-78583-7_3.
- Tien, Justin H. et al. (2020). "Online reactions to the 2017 'Unite the right' rally in Charlottesville: measuring polarization in Twitter networks using media follower-ship". en. In: *Applied Network Science* 5.10. DOI: 10.1007/s41109-019-0223-3. URL: <https://doi.org/10.1007/s41109-019-0223-3>.
- United Nations (2019). "United Nations Strategy and Plan of Action on Hate Speech". en. In: New York: United Nations. URL: https://www.un.org/en/genocideprevention/documents/advising-and-mobilizing/Action_plan_on_hate_speech_EN.pdf (visited on 12/27/2025).
- Vidgen, Bertie and Leon Derczynski (2020). "Directions in abusive language training data, a systematic review: Garbage in, garbage out". en. In: *PLOS ONE* 15.12, e0243300. DOI: 10.1371/journal.pone.0243300. URL: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0243300>.
- Vidgen, Bertie et al. (Aug. 2019). "Challenges and frontiers in abusive content detection". en. In: *Proceedings of the Third Workshop on Abusive Language Online*, pp. 80–93. DOI: 10.18653/v1/W19-3509. URL: <https://aclanthology.org/W19-3509/>.
- Wulczyn, Ellery, Nithum Thain, and Lucas Dixon (2017). "Ex Machina: Personal Attacks Seen at Scale". en. In: *Proceedings of the 26th International Conference on World Wide Web*, pp. 1391–1399. DOI: 10.1145/3038912.3052591. URL: <https://doi.org/10.1145/3038912.3052591> (visited on 12/27/2025).
- Zampieri, Marcos et al. (June 2019). "SemEval-2019 Task 6: Identifying and categorizing offensive language in social media (OffensEval)". en. In: *Proceedings of the 13th International Workshop on Semantic Evaluation (SemEval 2019)*, pp. 75–86. DOI: 10.18653/v1/S19-2010. URL: <https://doi.org/10.18653/v1/S19-2010> (visited on 12/27/2025).